

Wilfried Ariel Mulyawan | 陳順福

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EDUCATION

Engineering Management, Bandung Institute of Technology, Bandung, Indonesia

2020 – 2025

International Advanced Technology Program, National Taiwan University of Science and Technology, Taipei, Taiwan

Aug 2023 – Dec 2023

RESEARCH EXPERIENCE

AI Research Engineer, Singapore Management University, Hybrid (Full-time)

Feb 2025 – Sep 2025

- Built a 1,000+ question multimodal dataset to benchmark Vision-Language Models on ASEAN cultural reasoning, handling data collection, annotation, and validation under Professor Ngo Chong Wah and the SMU Multimedia Research Team.
- Co-authored “Seeing Culture: A Benchmark for Visual Reasoning and Grounding,” accepted at EMNLP 2025 Main Conference, the first benchmark that stress-tests VLMs on visually grounded reasoning across nine Southeast Asian cultures.
- Built a teacher-facing grading platform on top of multimodal LLMs that auto-evaluates student submissions and surfaces learning gaps, cutting grading time and giving students faster, more useful feedback.
- Ran evaluation cycles on a multilingual model to improve cultural relevance, digging into failure cases and reporting weekly findings on where current models break down, which fed directly into the next training iteration.

Lab Assistant, Lab of Innovation and Corporate System Development, ITB

Apr 2023 – May 2024

- Built an adaptive crypto trading system on a self-hosted server, training XGBoost and other scikit-learn models on real-time market data with automated re-learning cycles that sharpened entry and exit timing on leveraged positions.
- Wrote an undergraduate thesis applying machine learning to LQ45 trading, identifying the most predictive features behind returns across Indonesia’s top 45 stocks on Bursa Efek Indonesia.
- Studied how consumers react to AI-generated marketing versus human-made content using PLS-SEM and SEM-MGA, weighing the upsides and downsides of using AI as the creative backbone across two industry case studies.
- Taught independent classes in valuation, financial modelling, and risk assessment, and supported faculty in finance and economics courses through quiz design, grading, and student mentoring.

Research Intern, CITI Lab, NTUST (Taiwan Tech), Taipei, Taiwan

Sep 2023 – Dec 2023

- Took over a graduating master’s student’s Outcome Economy project under Professor Shuo-Yan Chou, building predictive maintenance and financial models for subscription-based smart home ownership.
- Modelled opportunity cost and inventory tradeoffs for the lab’s RMFS warehouse robotics project using the Newsboy framework, with the analysis folded into the team’s broader logistics pipeline.
- Co-developed demand forecasting models using the Northern Goshawk Optimization approach, running experiments and visualizations that were rolled into the lab’s downstream publications and industry partner reports.

INDUSTRY EXPERIENCE

Undisclosed Prediction Market Hedge Fund (Web3 / On-chain), Remote (Full-time)

Dec 2025 – Present

Managing Director Feb 2026 – Present

- Lead investment strategy and fund operations across a seven-figure USD volume portfolio, owning portfolio construction, position sizing, and risk frameworks across multiple systematic strategies that currently run at roughly 6% weekly ROI.
- Build quant strategies grounded in Avellaneda-Stoikov market making, Kelly criterion sizing, Black-Scholes pricing, Markov chains, Monte Carlo simulation, and Sharpe and Sortino-based risk evaluation.
- Designed the firm’s core trading infrastructure end to end: real-time on-chain data pipelines, AI agent and multi-agent execution layers, dashboard, and configuration and risk controls with a strong focus on clean execution and avoiding slippage.
- Run a small, cross-institutional team with collaborators from NYU, NTU, and HKU, covering the full research-to-production loop from alpha hypothesis to live deployment. Deploying capital and brainstorm ideas for strategy generated by collaborators

Portfolio Manager Dec 2025 – Feb 2026

- Designed and built systematic strategies with low pairwise correlation to capture alpha across distinct sectors of the on-chain prediction market, tuning win rate and risk-reward ratio so each strategy stayed proportional to the overall risk profile.

Assistant Analyst, Indonesia Ministry of Public Works and Housing, (Part-time)

Mar – Jun 2024

- Audited Indonesian regulations affecting the ministry, mapping how legislative changes shaped policy, decision-making, and day-to-day operating procedures across the strategic planning division.
- Analysed the ARO matrix organisational structure, surfacing workflow bottlenecks and recommending realignments so every team understood their scope and what they actually owned.
- Drafted revised business models by pulling apart existing legal and regulatory texts and spotting what could be improved, then folding those insights into new service delivery and stakeholder engagement plans.
- Worked directly with senior officials and external policy advisors to redefine team roles and responsibilities in line with reform objectives and legal constraints.

Digital Operations Manager, Radian Jaya, Indonesia (Hybrid)

Sep 2020 – June 2025

- Run daily operations of the business and online store, covering inventory, fulfilment, customer experience, and creative content across all sales channels, and wrote a lean returns and operations playbook that cut friction on customer issues.
- Built a data-driven pricing and barbell strategy spanning budget and premium tiers, paired with targeted ads, SEO, and paid social campaigns that grew annual sales by roughly 20% while lifting organic traffic.

LEADERSHIP & ENTREPRENEURSHIP

Co-Founder and COO, Ollifant Bebe (Babycare Brand), Indonesia

Oct 2024 – Present

- Co-founded a baby and kids care brand, running consumer research with parents on hair care needs, preferences, and pain points, and using my online-store background to drive organic reach and views for the brand from day one.
- Run day-to-day operations across product development, supplier coordination, branding, and go-to-market, taking new baby hair care products from concept to shelf and selling 150+ units in the first month of launch.

IISMA Awardee, Indonesian Government Student Exchange, NTUST, Taipei

Aug – Dec 2023

- Selected as 1 of 14 awardees from over 12,000 national applicants for the IISMA scholarship at NTUST, Taiwan, representing Indonesia in a flagship cross-cultural exchange program.
- Organized cultural events introducing Indonesian heritage to international students, and contributed to the campus Tennis and Table Tennis clubs throughout the semester abroad.

Founder and CEO, Miarsa (Student Housing Review Platform), Indonesia

2021 – 2022

- Founded a community-driven housing platform during COVID to fix what listing sites like Mamikos and Airbnb-style apps left out: real student experiences of the places they actually live in across multiple Indonesian cities.
- Led on-the-ground research interviewing residents on budget, day-to-day living conditions, guest policies, safety, and landlord behavior, turning raw interviews into structured listings students could trust.
- Grew the platform to 150+ active student users who relied on Miarsa to pick their housing, validating product-market fit through repeat use and word-of-mouth referrals across campuses.
- Owned the full stack of a 0-to-1 startup as founder: product design, content operations, user research, marketing, and direct community engagement.

PUBLICATIONS

- Seeing Culture: A Benchmark for Visual Reasoning and Grounding. EMNLP 2025 Main Conference.
- Designing LQ45 Stock Trading Strategy Using Machine Learning. ITB Final Thesis.
- The Impact of AI-Driven Visual Content in Marketing on Consumer Response. ITB LIPSP Lab.
- New Approach to Smart Home Subscription Systems in the Outcome Economy Era. NTUST CITI Lab.
- RMFS Opportunity Cost on Inventory Management. NTUST CITI Lab.

TECHNICAL SKILLS

- Machine Learning: Supervised/unsupervised learning, XGBoost, ensemble methods, time-series forecasting, statistical modeling (Negative Binomial, log-normal), regime detection, Quarter-Kelly position sizing, feature importance analysis, NLP, prompt engineering, model evaluation, benchmark construction.
- AI and Deep Learning: Large Language Models, Vision-Language Models, multimodal reasoning, multi-agent system design and orchestration, RAG pipelines, dataset construction, annotation pipelines.
- Data Science: Algorithmic trading system development, CLOB execution layer, Web3 Broker API integration, automated data collection, on-chain analytics, real-time market data pipeline, quantitative trading, financial modeling, and sentiment analysis.
- HCI and Systems Engineering: Interactive dashboard design, educational workflow AI integration, user-facing system design, self-hosted server deployment, end-to-end ML deployment, automated data pipelines.
- Programming and Tools: Python, PyTorch, scikit-learn, Pandas, SQL, NumPy, React, TypeScript, Vercel, web development, end-to-end ML deployment.

HONORS & AWARDS

- EMNLP 2025 Main Conference: co-author of “Seeing Culture”, a top-tier NLP venue acceptance.
- IISMA Scholarship: 1 of 14 selected from over 12,000 national applicants (Indonesian Government, 2023).
- PMW Entrepreneurship Award and national startup funding for Ticketoad, a secondhand event ticket marketplace tackling scalping and unsafe peer-to-peer resale (Program Mahasiswa Wirausaha, 2022).
- 2nd Place Regional Physics (2018) and 3rd Place Regional Astronomy (2017), National Science Olympiad, DKI Jakarta.
- Silver Medalist in Chess, National Sports Week, Indonesia (2014); 4th Place Regional Table Tennis, National Sports Olympiad (2013).

LANGUAGES AND STANDARDIZED TESTS

English: IELTS Band 7 | Duolingo English Test: 140 (equivalent to IELTS 7.5) Mandarin: HSK 3 Indonesian: Native